



The WHO (World Health Organization) met for the first time in 1948. This UN (United Nations) agency has played a part in promoting healthcare and eradicating diseases.

1949

Naming the "Big Bang"

British astronomer Sir Fred Hoyle coined the term "Big Bang" (see pp.182–183) to describe the theory that the Universe began by expanding from a single point. He used the term for the first time on a BBC radio program.

1948

Bird rediscovered

Living examples of the flightless takahē bird were discovered near Lake Te Anau, in New Zealand's South Island, by Dr. Geoffrey Orbell. The 24.8-in- (63-cm-) long bird was last sighted in 1898 and was thought to be extinct.



1949

Positive effects of cortisone

American physician Philip S. Hench discovered how a hormone (chemical in the body) called cortisone could reduce inflammation in patients suffering from the disease rheumatoid arthritis. In the same year, American chemist Percy L. Julian devised a quick and affordable method of making cortisone in laboratories to meet the growing demand for it.

1949

Structure of penicillin

British chemist Dorothy Crowfoot (later Hodgkin) and her team published the molecular structure of penicillin, a group of antibiotics (medication against bacterial infections). X-ray crystallography (a process in which the patterns cast by reflected X-rays are analyzed) was used to make the map of penicillin's atoms and bonds. This helped to develop more successful antibiotics to fight resistant bacteria.

Molecular model of penicillin

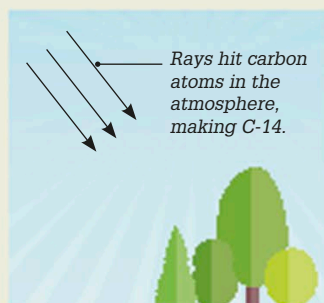


1950

1949 RADIOCARBON DATING

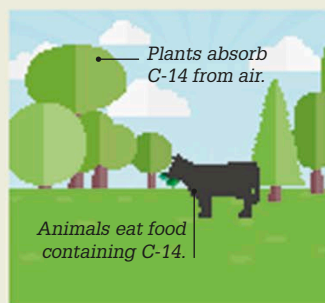
American chemists James Arnold and Willard Libby developed radiocarbon dating. Organic materials (made from living things) contain normal carbon, called carbon-12 (C-12), and a radioactive isotope (form) called carbon-14 (C-14). While C-12 levels stay the same, C-14 decays at a known rate (halving in quantity every 5,730 years)—so the ratio of C-14 to C-12 decreases with age. By measuring this ratio, scientists can find out the age of ancient objects made with organic materials, such as wood and cotton.

Taking a sample from a skull for radiocarbon dating



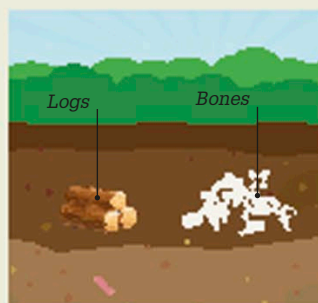
1. C-14 is formed

Cosmic rays collide with atoms in the upper atmosphere and produce C-14, which has two more neutrons than regular carbon.



2. Ingesting C-14

Plants absorb C-14 from the air, and animals and humans obtain C-14 through food (plants and animals) that they eat.



3. Death and decay

Following death and burial of an organism, the C-14 in it decays at a constant rate and its amount in the object decreases.



4. Dating samples

Measuring the amount of C-14 in a sample gives an accurate age of objects up to 50,000 years old.

The amount of C-14 is high in fresh remains.

The amount of C-14 is low in fossils.